



**NITTE**  
(Deemed to be University)

**NMAM INSTITUTE  
OF TECHNOLOGY**

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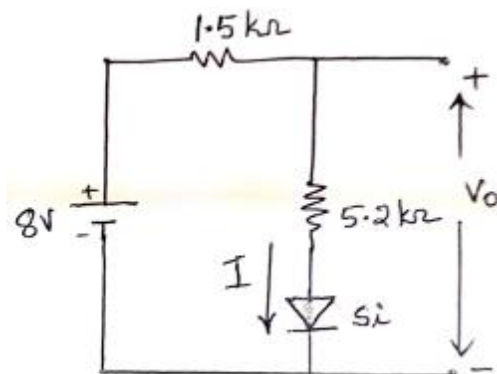
## QUESTION BANK

**Subject: Basic Electronics**

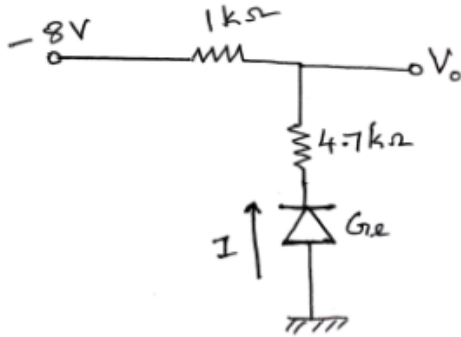
**Course Code: EC1001-1 (3 credits)**

### UNIT I

1. Draw the VI characteristics for Silicon and Germanium diodes and define the diode parameters.
2. With neat circuit diagram and input output waveforms, explain the working of a half wave rectifier. Derive all the expressions relevant to current and voltages with efficiency and ripple factor.
3. With neat circuit diagrams and relevant graphs, explain the diode models.
4. With neat circuit diagram and input output waveforms, explain the working of a bridge rectifier. Derive all the expressions relevant to current and voltages with efficiency and ripple factor.
5. For the diode circuit shown in figure below, calculate the current  $I$  and the output voltage  $V_o$ .



6. For the diode circuit shown in figure below, calculate the current  $I$  and output voltage  $V_o$ .



7. A sinusoidal voltage of peak value 40 V and frequency 50 Hz is applied to a HWR without filter. It has a load of  $1\text{ k}\Omega$  with forward resistance of the diode as  $8\Omega$ . Calculate
  - a) Peak, dc and rms value of load current.
  - b) DC output power
  - c) AC input power and
  - d) Rectifier efficiency.
8. A full wave rectifier has 4 diodes with RMS voltage of 110 V. Forward resistance of each diode is  $25\Omega$  with a load resistance of  $1\text{ k}\Omega$ . Calculate
  - i) Maximum value of current
  - ii) DC value of the current through load
  - iii) DC value of the voltage across the load.
9. Draw the VI characteristics of a Zener diode and define the Zener parameters.
10. A 24 V, 600 mW Zener diode is used for providing a stabilized voltage to a variable load from a 32 V power supply. Calculate the
  - i) Value of series resistance required in the circuit.
  - ii) Zener current when the load is  $1200\Omega$
11. Design a 6.2 V Zener voltage regulator to operate from a 16 V power supply, so as to supply maximum possible load current with a minimum Zener current  $I_{z\min} = 14.5\text{ mA}$ . Assume  $P_{z(\max)} = 400\text{ mW}$ . Also calculate the value of the load  $R_L$  to be connected in the circuit.
12. A 12 V, 360 mW Zener diode is used for providing a stabilized voltage to a variable load from a 24 V power supply. Calculate the
  - i) Value of series resistance required.
  - ii) Zener current when the load is  $1240\Omega$

13. Define line regulation and load regulation in Zener voltage regulators.
14. Bring out the relationship between the transistor current gains  $\alpha$  and  $\beta$ .
15. With neat circuit diagrams explain the CB configuration of BJT and draw the input/output characteristics.
16. With neat circuit diagrams explain the CE configuration of BJT and draw the input/output characteristics.
17. With a neat circuit figure and a graph, show the DC load line analysis of a BJT.
18. A BJT has  $I_C = 3 \text{ mA}$  and  $I_E = 3.03 \text{ mA}$ . Compute the value of  $\beta$  of the transistor used. Assuming no change in the base current, find the new collector current when the transistor is replaced by a new one with  $\beta$  of 70.
19. Determine  $\alpha$  and  $\beta$  of a transistor for  $I_C = 1 \text{ mA}$  and  $I_B = 25 \mu\text{A}$ . determine the new base current to give  $I_C = 5 \text{ mA}$ .
20. Calculate  $\alpha$ ,  $\beta$  and  $I_B$  of a transistor that has  $I_C = 2.5 \text{ mA}$  and  $I_E = 2.55 \text{ mA}$ .
21. With a neat circuit figure and a graph, show the DC load line analysis of a BJT.
22. With a neat circuit diagram, explain how BJT can be made to operate as a switch.
23. With a neat circuit diagram of a CE-RC coupled amplifier,
  - i) Describe the significance of each component in the circuit
  - ii) Illustrate the phase reversal concept
  - iii) Draw the input output waveforms
24. With the neat connection diagrams, explain the drain and transfer characteristics of n-channel JFET.
25. With the neat connection diagrams, explain the drain and transfer characteristics of n-channel enhancement type MOSFET.
26. Explain the working of a CMOS inverter with a neat circuit diagram.
27. For a n-channel JFET,  $I_{DSS} = 6 \text{ mA}$ ,  $V_P = -4.5 \text{ V}$ . Determine  $I_D$  at  $V_{GS} = -2 \text{ V}$  and  $V_{GS} = -6 \text{ V}$
28. For an n-channel JFET,  $I_{DSS} = 8 \text{ mA}$  and  $V_P = -4 \text{ V}$ . What is the value of  $I_D$  for
  - (i)  $V_{GS} = -1 \text{ V}$  and
  - (ii)  $V_{GS} = -2 \text{ V}$ .
29. For an n-channel JFET,  $I_{DSS} = 6 \text{ mA}$  and  $V_P = -4.5 \text{ V}$ . What is the value of  $I_D$  for  $V_{GS} = -6 \text{ V}$ .

